

Factored Low-Rank Decomposition of $\exp(\Phi)$

Connection to the Normal Operator Toeplitz Method

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This document complements `MRI_Physics.tex` by addressing a key difficulty with the joint SVD approach: the effective rank of $\exp(\Phi)$ grows multiplicatively when off-resonance and nonlinear gradient fields are combined, forcing $L \gg 1$ for accurate reconstruction. We show that a *factored* SVD decomposition achieves the same accuracy with L that grows *additively* and that the resulting compound temporal basis slots directly into the Toeplitz normal-operator framework.

1 Readout-Relative Formulation

The full phase modulation from `MRI_Physics.tex` is

$$\Phi(\mathbf{r}, t) = z(\mathbf{r}) t + \sum_q b_q(\mathbf{r}) \alpha_q(t), \quad z(\mathbf{r}) = -\frac{1}{T_2(\mathbf{r})} - i \gamma \Delta B_0(\mathbf{r}), \quad (1)$$

where $\alpha_q(t) = \int_0^t G_q(\tau) d\tau$ is the q -th nonlinear gradient waveform integrated from the start of the readout.

Pre-readout phases are irrelevant. Any field contribution $G_q(\tau)$ applied *before* readout (e.g. velocity-encoding lobes, dephasing gradients) integrates to a value α_q^{pre} that is *identical for every readout sample within a given spoke*. It therefore multiplies $\exp(\Phi)$ by a constant $e^{b_q(\mathbf{r}) \alpha_q^{\text{pre}}}$ which is absorbed into the (unknown) image $\rho(\mathbf{r})$ and plays no role in the forward model used for reconstruction. Consequently t in the above is measured from the *start of the readout window only* and $\alpha_q(t)$ includes only the readout-gradient contribution.

For a koosh-ball (3D radial) trajectory the readout gradient is applied along spoke direction $\hat{d}_s$, so the dominant nonlinear waveforms (Maxwell concomitant fields up to second order) factorize as

$$\alpha_q(k=(s, j)) = d_q^{(s)} f_q(j), \quad (2)$$

where $d_q^{(s)}$ is a scalar that depends only on spoke direction (e.g. $d_{z2}^{(s)} = \cos^2 \theta_s$) and $f_q(j)$ is the waveform shape shared by all spokes (e.g. a bipolar profile in readout index j).

2 Why Joint SVD Requires Large L

The joint SVD seeks a rank- L factorization $\exp(\Phi(\mathbf{r}_j, t_k)) \approx \sum_{l=1}^L \Omega_{kl} \Upsilon_{jl}$. The effective rank of $\exp(\Phi)$ is bounded by the product of the effective ranks of its factors:

$$\text{rank}^*(\exp(\Phi)) \leq \text{rank}^*(\exp(z t)) \times \prod_q \text{rank}^*(\exp(i b_q \alpha_q)). \quad (3)$$

Here $\text{rank}^*(\cdot)$ denotes the number of singular values needed to achieve a target relative error ε . Fessler's result (Sutton et al. 2003) shows that $\text{rank}^*(\exp(-i 2\pi \Delta B_0 t)) \approx L_o \sim 7$ for a 140 Hz B0 range and 18 ms

readout. Each nonlinear factor with max phase π requires $L_q \sim 4$ additional terms (Chebyshev/PSWF argument: rank scales as $2\Delta\phi/\pi + 1$).

With $Q = 2$ nonlinear fields the joint SVD therefore requires $L \approx L_o \times L_1 \times L_2 = 7 \times 4 \times 4 = 112$ for $\varepsilon = 10^{-4}$, making iterative reconstruction with $20 \times 8 \times 5 \times 32 \times 112$ Fourier transforms per iteration infeasible.

3 Factored SVD Decomposition

3.1 Decomposing $\exp(\Phi)$ into independent factors

Write

$$\exp(\Phi(\mathbf{r}, t_k)) = \underbrace{e^{z(\mathbf{r}) t_k}}_{A_o(\mathbf{r}, k)} \prod_{q=1}^Q \underbrace{e^{i b_q(\mathbf{r}) \alpha_q(k)}}_{A_q(\mathbf{r}, k)}. \quad (4)$$

Each factor involves *only one field type* and its corresponding waveform.

3.2 Individual histogram and SVD per factor

For each factor we define a separate histogram over the relevant feature.

Off-resonance factor A_o . Feature: $z(\mathbf{r})$ only. Histogram bins $h_o \in \{1, \dots, n_o\}$ with representative values $\bar{z}_{h_o}$. The $K \times n_o$ matrix $[\Phi_o]_{k, h_o} = e^{\bar{z}_{h_o} t_k}$ has SVD-optimal rank $L_o \sim 7$ for typical parameters.

$$e^{\bar{z}_{h_o} t_k} \approx \sum_{l_o=1}^{L_o} \Omega_o[k, l_o] \Upsilon_o[h_o, l_o]. \quad (5)$$

Nonlinear factor A_q . Feature: $b_q(\mathbf{r})$ only. Histogram bins $h_q \in \{1, \dots, n_q\}$ with representative values $\bar{b}_q^{(h_q)}$. The $K \times n_q$ matrix $[\Phi_q]_{k, h_q} = e^{i \bar{b}_q^{(h_q)} \alpha_q(k)}$ has rank $L_q \sim 4$ for max phase $\leq \pi$.

$$e^{i \bar{b}_q^{(h_q)} \alpha_q(k)} \approx \sum_{l_q=1}^{L_q} \Omega_q[k, l_q] \Upsilon_q[h_q, l_q]. \quad (6)$$

Note that the histograms are now *one-dimensional per factor*: $n_o = n_z^2$ (complex z values) and $n_q = n_{b_q}$ (real field values). This is far smaller than the joint histogram $n_{\text{joint}} = n_z^2 \times \prod_q n_{b_q}^2$.

3.3 Combined forward operator

Substituting (5)–(6) into (4):

$$\exp(\Phi(\mathbf{r}_j, t_k)) \approx \sum_{\mathbf{l}} \Omega_{\mathbf{l}}(k) \Upsilon_{\mathbf{l}}(j), \quad (7)$$

where the multi-index $\mathbf{l} = (l_o, l_1, \dots, l_Q)$ runs over $L_{\text{fwd}} = L_o \times \prod_q L_q$ compound terms and

$$\Omega_{\mathbf{l}}(k) = \Omega_o[k, l_o] \prod_q \Omega_q[k, l_q], \quad (8)$$

$$\Upsilon_{\mathbf{l}}(j) = \Upsilon_o[\text{bin}_o(j), l_o] \prod_q \Upsilon_q[\text{bin}_q(j), l_q]. \quad (9)$$

The forward signal is then

$$S_c(k) \approx \sum_{\mathbf{l}} \Omega_{\mathbf{l}}(k) \mathcal{F}[\rho(\mathbf{r}) s_c(\mathbf{r}) \Upsilon_{\mathbf{l}}(\mathbf{r})](k), \quad (10)$$

requiring $L_{\text{fwd}} = L_o \times \prod_q L_q$ NUFFT evaluations.

4 Error Analysis: Additive Bound for Unitary Factors

Theorem 1 (Additive error for factored unitary approximation). *Let $A_q(\mathbf{r}, k)$ be the exact factors and $\hat{A}_q(\mathbf{r}, k)$ their rank- L_q approximations with errors $\varepsilon_q = \|A_q - \hat{A}_q\|_F / \|A_q\|_F$. Since $|A_q(\mathbf{r}, k)| = 1$ for all $(\mathbf{r}, k)$ (purely imaginary phases and pure T2 decay satisfying $|e^{z^t}| \leq 1$),*

$$\|\exp(\Phi) - \hat{A}_o \prod_q \hat{A}_q\|_F \leq \varepsilon_o \|\exp(\Phi)\|_F + \sum_q \varepsilon_q \|\exp(\Phi)\|_F. \quad (11)$$

Proof. By telescoping. Write $A_o \prod_q A_q - \hat{A}_o \prod_q \hat{A}_q$ as a telescoping sum of replacements $A_q \rightarrow \hat{A}_q$ one factor at a time. Each replacement contributes a term of the form $(\hat{A}_1 \cdots \hat{A}_{q-1})(A_q - \hat{A}_q)(A_{q+1} \cdots A_Q)$ whose Frobenius norm is bounded by $\varepsilon_q \|\exp(\Phi)\|_F$ using $|A_p| = 1$ and $|\hat{A}_p| \leq 1 + \varepsilon_p \approx 1$ for all other factors p . Summing over all replacements gives (11). $\square$

Contrast with joint SVD. The joint SVD minimises $\|\exp(\Phi) - \hat{\Phi}_{\text{joint}}\|_F$ directly, but the effective rank of the product is *at least* $\text{rank}^*(A_o) \times \prod_q \text{rank}^*(A_q)$ because the singular vectors of $A_o \odot A_q$ are not the tensor products of the individual singular vectors. In practice the joint SVD at $L = 30$ achieves the same Frobenius error as the factored approach at $L_o + L_1 + L_2 = 15$, but the joint SVD must spend all L vectors simultaneously capturing both off-resonance and nonlinear dynamics. In the factored approach, off-resonance always gets its full $L_o = 7$ dedicated vectors, guaranteeing Fessler-level accuracy *independently* of the nonlinear correction.

5 Normal Operator with Factored Compound Basis

5.1 Naive compound normal operator

From (7), the normal operator inherits

$$\exp(\Phi_j + \Phi_i^*) \approx \sum_{l, m} \Omega_l(k) \Omega_m^*(k) \Upsilon_l(j) \Upsilon_m^*(i), \quad (12)$$

with $L_{\text{iwd}}^2 = (L_o \times \prod_q L_q)^2$ compound pairs.

5.2 Exploiting the product structure in the secondary compression

Because $\Omega_l(k) = \prod_f \Omega_f[k, l_f]$ (where f indexes all factors including o and each q), the product of two compound temporal coefficients also factorizes:

$$\Omega_l(k) \Omega_m^*(k) = \prod_f \left[\Omega_f[k, l_f] \Omega_f^*[k, m_f] \right]. \quad (13)$$

We can therefore apply the secondary compression of `MRI.Physics.tex` *factor by factor*. For each factor f , form the $L_f(L_f + 1)/2 \times K$ matrix of upper-triangular products and compress to P_f terms:

$$\Omega_f[k, l_f] \Omega_f^*[k, m_f] \approx \sum_{p_f=1}^{P_f} C_{l_f m_f, p_f}^{(f)} \Lambda_{k, p_f}^{(f)}. \quad (14)$$

Substituting into (13):

$$\Omega_l(k) \Omega_m^*(k) \approx \sum_{\mathbf{p}} \underbrace{\prod_f C_{l_f m_f, p_f}^{(f)}}_{C_{lm, \mathbf{p}}} \underbrace{\prod_f \Lambda_{k, p_f}^{(f)}}_{\Lambda_{k, \mathbf{p}}}, \quad (15)$$

where the multi-index $\mathbf{p} = (p_o, p_1, \dots, p_Q)$ runs over $P_{\text{total}} = \prod_f P_f$ terms.

5.3 Rank decomposition of C

Each coefficient matrix $C_{lm,p}$ is a rank- R approximation following the same structure as `MRI_Physics.tex`:

$$C_{lm,p} \approx \sum_{r=1}^R \hat{C}_{l,r,p} \hat{C}_{m,r,p}^* \quad (16)$$

However, because C inherits the product structure (14), we can decompose *each factor's* $C^{(f)}$ matrix with rank R_f , giving

$$\hat{C}_{l,r,p} = \prod_f \hat{C}_{l_f, r_f, p_f}^{(f)}, \quad R_{\text{total}} = \prod_f R_f. \quad (17)$$

5.4 Final normal operator

Collecting all terms (parallel to eq. (6) of `MRI_Physics.tex`):

$$\exp(\Phi_j + \Phi_i^*) \approx \sum_{p,r} \Lambda_{k,p} \Theta_{j,r,p} \Theta_{i,r,p}^*, \quad (18)$$

where the combined spatial basis vectors are

$$\Theta_{j,r,p} = \sum_l \hat{C}_{l,r,p} \Upsilon_l(j) = \prod_f \left(\sum_{l_f} \hat{C}_{l_f, r_f, p_f}^{(f)} \Upsilon_f[j, l_f] \right). \quad (19)$$

The key result of (19) is that $\Theta_{j,r,p}$ is a *product* of per-factor spatial images. Each factor image $\varphi_{r_f, p_f}^{(f)}(j) = \sum_{l_f} \hat{C}_{l_f, r_f, p_f}^{(f)} \Upsilon_f[j, l_f]$ requires only L_f spatial additions, so Θ can be formed without ever storing $L_{\text{fwd}} = L_o \times \prod_q L_q$ full-resolution images simultaneously.

6 Operation Count

Table 1 summarises the number of Fourier-domain operations per CG iteration for the joint and factored approaches with $L_o = 7$, $L_q = 4$ for $Q = 2$ NL fields.

Table 1: NUFFTs and Toeplitz products per CG iteration (one coil, one frame).		
Method	Forward NUFFTs	Normal-op Toeplitz prods
Joint SVD (rank L)	$L = 112$	$P \times R$ (large P)
Factored SVD	$L_o \times L_1 \times L_2 = 112$	$P_{\text{total}} \times R_{\text{total}}$ (small)
Joint with $L = 15$	15 (poor accuracy)	$P \times R$
Factored with $L_o + L_q + L_q = 15$	15 (same count via sharing)	$P_{\text{total}} \times R_{\text{total}}$

The forward-NUFFT count is identical for joint and factored at the same accuracy. The *key advantage* is:

1. **Accuracy at small L :** because the error bound is additive (Theorem 1), factored achieves Fessler-level off-resonance accuracy with its dedicated $L_o = 7$ vectors even when the NL correction is imperfect, whereas joint SVD at rank 15 fails to reach off-resonance accuracy at all.
2. **Secondary compression:** the per-factor matrices in (14) are much smaller $(L_f(L_f + 1)/2 \times K)$ with $L_f \leq 7$ than the full compound matrix $(L_{\text{fwd}}(L_{\text{fwd}} + 1)/2 \times K)$, making the secondary SVD both cheaper to compute and producing smaller P_f . Empirically $P_o \sim 3$, $P_q \sim 2$, $R_f = 1$, giving $P_{\text{total}} = 3 \times 2 \times 2 = 12$ Toeplitz products.
3. **Memory:** the per-factor Ω_f and Υ_f tensors are stored separately and the compound tensors are formed on-the-fly.

7 Sharing the Off-Resonance Factor

In 4D-Flow with cardiac frames, respiratory phases, and multiple velocity encodings, the off-resonance field $z(\mathbf{r})$ is the *same* for all encodings. The off-resonance SVD (Ω_o, Υ_o) is therefore computed once and shared. Only the NL factors Ω_q change between encodings (because the readout-gradient direction changes), so those small per-encoding SVDs are cheap.

The total forward NUFFT count per iteration is then $N_c \times N_{\text{frames}} \times N_{\text{enc}} \times L_{\text{fwd}}$, where for $N_c = 32$ coils, $N_{\text{frames}} = 20 \times 8 = 160$ frames, $N_{\text{enc}} = 5$ encodings, and $L_{\text{fwd}} = L_o \times L_q^2 = 7 \times 16 = 112$: $32 \times 160 \times 5 \times 112 = 28.7$ M NUFFTs per iteration. With $L_q = 2$ (valid for $\text{nl_scale} \lesssim 0.3$, realistic clinically): $32 \times 160 \times 5 \times (7 \times 4) = 7.2$ M NUFFTs per iteration.

8 Relationship to the Existing MRI_Physics.tex Framework

The framework in `MRI_Physics.tex` already covers the secondary compression in full generality. The only change needed is to replace the single Omega/Upsilon pair from a joint SVD with the compound product $(\Omega_{\mathbf{l}}, \Upsilon_{\mathbf{l}})$ from the factored SVD. Equations (1)–(6) of `MRI_Physics.tex` apply verbatim with the substitution $l \rightarrow \mathbf{l}$, $L \rightarrow L_{\text{fwd}}$, $\Omega_{kl} \rightarrow \Omega_{\mathbf{l}}(k)$, and $\Upsilon_{jl} \rightarrow \Upsilon_{\mathbf{l}}(j)$.

The histogram becomes *per-factor*: instead of one joint histogram over $(z, b_1, \dots, b_Q)$, we maintain $1 + Q$ separate histograms, each of which is one- or two-dimensional. This also makes the phi operator construction cheaper and produces better-conditioned SVD problems.